Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/10/2019
PI Name:	Rutkove, Seward M.D.		
Project Title:	Musculoskeletal Response to a Partial-gravity Analog in Rats: Structural, Functional, and Molecular Alterations		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Artificial Gravity. Appendix D: NASA HRP Artificial Gravity Opportunity
Start Date:	06/09/2016	**End Date:**	01/01/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: Extended to 1/1/2020 per NSSC information (Ed., 9/17/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 3/11/19) NOTE: Extended to 5/31/2019 per NSSC information (Ed., 10/23/18) NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 12/6/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center/Harvard Medical Center)		
Grant/Contract No.:	NNX16AL36G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Health of the musculoskeletal system depends on exposure to mechanical loading. Removal or reduction of mechanical loading precipitates muscle atrophy and bone loss. Thus, despite rigorous exercise protocols and other countermeasures, muscle atrophy and bone loss are still virtually inevitable consequences of spaceflight. One approach for ameliorating the development of these untoward consequences of space travel is by the induction of artificial gravity during spaceflight via centrifugation. The strength of the artificial gravity depends both on the speed of rotation and diameter of the ship. While it is possible to achieve centrifugal force equivalent 1.0 G, it is also possible that lower levels of force may also be effective for preventing muscle atrophy and irreversible bone loss. In this study, we propose to evaluate the potential effectiveness of artificial gravity for the prevention of muscle atrophy and weakness and bone loss by studying rats using an innovative system for partial unloading that has been developed previously for mice by members of our team. We will study the musculoskeletal response of adult male rats (n=14 / group) following exposure to different degrees of unloading representing 0.2 G, 0.4 G, and 0.7 G as well as fully loaded animals for 1, 2, and 4 weeks. We will study the impact of this partial gravity analog environment on muscle and bone structure and function, including performing histological analyses, in situ force measurement of muscle contraction, as well as measurement of bone mass, microarchitecture, and strength. In addition, we will perform both serological and tissue molecular analyses to better understand the mechanisms and time course of development of these changes at different degrees of unloading. This work will also allow us to more effectively understand the relationship between the development of muscle atrophy and weakness and bone deterioration at varying levels of artificial gravity. At the conclusion of this study, we will have defined the potential impact of partial artificial gravity for prevention of musculoskeletal dysfunction and deterioration. This work will inform the future application of artificial gravity in human research and space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We believe that this research will positively impact in life on Earth since it will inform us about the effects of reduced muscle and bone use that occur normally in our daily lives and as a consequence of disease or injury. For example, using a cane may reduce the normal weightbearing of a single limb. While this obviously may be necessary from an assistive standpoint, this could lead to additional atrophy of the limb muscles and osteopenia of the bone.
Task Progress:	During the past year we completed the bulk of our originally planned work studying the musculoskeletal effects of partial unloading at different levels. We also have completed most of the analyses of bone and muscle and have submitted several manuscripts describing this work and are currently completing several of the of the RNAseq analyses. We have identified dose-depending effects of partial weigthbearing (also termed fractional gravity) on both muscle and bone health. These have included reductions in bone mineral density and reduced muscle mass and force production. We have also demonstrated a dose-dependent increase in relative number of fast-twitch fibers in the muscle that also follows a dose-dependent pattern. There remain several manuscripts in preparation from this work that we hope to have submitted in the coming months. In addition to this work, we extended our model to studies of resveratrol in preserving muscle function as well as looking at varying loading levels. These efforts also led to augmentation funding to study several other aspects of the model that require additional refinement, including evaluating stress levels in the rats, evaluating blood flow in the hind limbs, and assessing how equally or unequally the four limbs are unloaded.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Mortreux M, Ko FC, Riveros D, Bouxsein ML, Rutkove SB. "A Novel Rat Partial Gravity Analogue for Longitudinal Investigations: Musculoskeletal Alterations." Presented at the podium and during a poster session at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Ko FC, Mortreux M, Riveros D, Nagy JA, Rutkove SB, Bouxsein ML. "Dose-dependent skeletal deficits in rats by varied reductions in mechanical loading." Presented at the Human Reseach Program Investigators Workshop: Human Exploration and Discovery: the Moon, Mars and Beyond!, 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Mortreux M, Riveros D, Bouxsein ML, Rutkove SB. "A Moderate Daily Dose of Resveratrol Mitigates Muscle Deconditioning in Rats Exposed to a Martian Gravity Analogue." Presented at the Human Research Program Investigators Workshop: Human Exploration and Discovery: the Moon, Mars and Beyond!, 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Ko FC, Mortreux M, Riveros D, Nagy JA, Rutkove SB, Bouxsein ML. "Dose-dependent skeletal deficits in rats by varied reductions in mechanical loading." Presented at the podium at the 2019 Orthopedic Research Society, Austin, Texas, February 2-5, 2019. 2019 Orthopedic Research Society, Austin, Texas, February 2-5, 2019. , Jan-2019
Articles in Other Journals or Periodicals	Mortreux M, Ko FC, Riveros D, Bouxsein ML, Rutkove SB. "Longitudinal Time Course of Muscle Impairments in a Rat Model of Partial Gravity." NPJ Microgravity (in revision) as of April 2019. , Apr-2019
Articles in Peer-reviewed Journals	Mortreux M, Riveros D, Bouxsein ML, Rutkove SB. "A moderate daily dose of resveratrol mitigates muscle deconditioning in a Martian gravity analog. " Front Physiol. 2019 Jul 18;10:899. eCollection 2019. https://doi.org/10.3389/fphys.2019.00899 ; PubMed PMID: 31379604; PubMed Central PMCID: PMC6656861 , Jul-2019
Articles in Peer-reviewed Journals	Mortreux M, Riveros D, Bouxsein ML, Rutkove SB. "Mimicking a space mission to Mars using hindlimb unloading and partial weight bearing in rats." Journal of Visualized Experiments (JoVE). 2019 Apr;146:e59327. Published online 4/4/2019. https://doi.org/10.3791/59327 ; PubMed PMID: 31009001 , Apr-2019